

# Sammy Abdella

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## EDUCATION

Columbia University in The City of New York  
B.S. Computer Science

Expected 05/2027

## SKILLS

**Technical:** Python, C++, Java, JavaScript, SQL, HTML/CSS, Ruby, OAuth2, JWT, Docker, Kubernetes  
**Languages:** French, Arabic, English

## EXPERIENCE

**Web Developer Intern | Columbia University IT Department | New York, NY** 01/2024 - 08/2024

- Migrated 43 Columbia departmental websites from Drupal 7 to Drupal 10, writing migration scripts, refactoring custom modules, building/refreshing site themes with HTML/CSS, running regression tests, improving average page load times by ~35%, and removing 100% of deprecated modules in production
- Integrated a Django, PostgreSQL, and RESTful API backend with a new Columbia admissions website, scaling reliably to 50,000+ applicants per cycle, cutting form submission errors by ~40%, and reducing manual data correction work by ~50%

**Backend Software Developer Intern | Zitouna Bank | Remote** 01/2024 – 05/2024

- Engineered RESTful APIs using Python/Django to cut response times by 30% and boost transaction throughput by 25%
- Improved PostgreSQL for real-time financial workloads by improving indexing and query plans; cut p95 latency ~35% and improved reliability under peak traffic
- Authored 50+ unit/integration tests for critical transaction endpoints and database logic; increased backend test coverage from 62% → 82% and reduced post-release defects by 25%.

**Research Intern | Columbia University AI Laboratory | New York, NY** 04/2023 - 09/2024

- Led the development for gAyl, an AI tool for LGBTQ+ suicide prevention, designed six model-based safety benchmarks to generate treatment plan recommendations and achieved <15% false-negative rate on unsafe-response detection across 1,000+ high-risk prompts
- Trained LLM-based classifiers in PyTorch and Hugging Face Transformers to automate safety evaluation, reducing manual benchmarking time by ~95%
- Publication (co-author): DOI: 10.7916/z6vc-d867

**Software Engineering Intern | Compyle.ai | San Francisco, California** 11/2023 - 05/2024

- Developed core product features for the web app (exam/workflow creation, session management, analytics) and shipped to production via code reviews and CI; improved user completion rate by ~15%.
- Integrated model-based summarization/Q&A into the product with guardrails (prompt templates, rate limiting, logging); reduced support tickets by ~25% and improved response consistency.

## Projects

**AI-Powered Exam Tutor | Python, Django, PostgreSQL, Redis, Celery, Docker** 11/2023 - 05/2024

- Built a web app that generates timed practice exams, stores attempts, and produces graded reports with explanations and targeted remediation suggestions.
- Deployed with Columbia University's Learning Center to support 50+ students across Data Structures, Advanced Programming, and Multivariable Calculus exam prep.